

Isotope and Solvent Effects of Deuterium on Aconitase¹

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The activity of aconitase has been compared in H₂O and in 99% D₂O, with deuterio-citrate and deuterioisocitrate as well as the protiated substrates, citrate, isocitrate, and aconitate. The rates of all six interconversions were markedly inhibited by D₂O; the conversion of aconitate to citrate was affected much more than the conversion of aconitate to isocitrate. The values obtained for Michaelis constants and maximum velocities satisfactorily predict equilibrium conditions in H₂O and D₂O. Negligible isotope effects were observed with deuteriocitrate or deuterioisocitrate; this observation is consistent with the hypothesis that the first (and rate-limiting) step in the dehydration of these compounds is the rupture of the C—O rather than the C—H bond.

Several years ago it was shown that the enzymic hydration of fumarate in D₂O took place at less than half the rate observed in ordinary water (1). The reverse reaction, the dehydration of malate, was inhibited by only 20% in D₂O. There was a slight effect on the apparent equilibrium constant.

This report is concerned with a similar study on another enzyme concerned with reversible hydrations, namely aconitase [citrate (isocitrate) hydro-lyase, E.C. 4.2.1.3] which catalyzes the interconversions among *cis*-aconitate, citrate, and threo-D_s-isocitrate (2) (referred to hereafter as (+) isocitrate).

Speyer and Dickman (3), studying 4 of the 6 interconversions in 93% D₂O, observed inhibitions of 35–60%. However, they did not measure the effects on maximum velocities or Michaelis constants. We have evaluated these kinetic parameters for both protiated and deuteriated substrates in ordinary water and in 99.6% D₂O.

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METHODS

Enzyme. Aconitase prepared from beef heart was used in the preliminary experiments. Most of the kinetic data reported here were obtained on preparations from beef liver. The purification procedures consisted of two ethanol fractionations and one ammonium sulfate fractionation (4). Further purification was not attempted; after dialysis of the enzyme, it must be reactivated with cysteine and ferrous ion, and the Fe⁺⁺-citrate and Fe⁺⁺-isocitrate complexes interfere with the spectrophotometric estimation of aconitate (5). The final protein precipitate was redissolved (about 10 mg per milliliter) in 0.125 M acetate buffer, pH 5.7, and stored at –10°. In most cases, portions of the precipitates were dissolved in acetate buffer in D₂O for use in experiments in that solvent in order to minimize dilution of deuterium by ordinary hydrogen (protium).

Substrates. Several samples of *cis*-aconitic anhydride were used in these experiments. Without exception, all solutions contained a small amount of the *trans* isomer, even after repeated crystallization of the anhydride from benzene. The presence of *trans*-aconitate was detected in the course of experiments in which equilibrium conditions were measured; with aconitate as the initial substrate, there was considerably higher absorption (240 mμ) at equilibrium than was observed with isocitrate and citrate. This extraneous

absorbing material was identified as *trans*-aconitate by paper chromatography with a solvent system of ethyl acetate, acetic acid, and water (6), and a pyridine-acetic anhydride spray (7). The R_F values for *cis*- and *trans*-aconitate were 0.60 and 0.89, respectively. In most of the experiments reported here, a sample of *cis*-aconitic anhydride purchased from Calbiochem was used. On the basis of the extra absorption at equilibrium, solutions of this material contained 6% *trans*-aconitate. As will be shown below, the *trans* isomer is a weak competitive inhibitor of aconitase, and its presence at this level has very little effect on either the Michaelis constant or the maximum velocity for *cis*-aconitate.

Stereospecifically labeled 3-deuterio-(+)-isocitrate and 2-deuteriocitrate were prepared enzymically from *cis*-aconitate in D_2O ; after equilibrium was reached, the reaction was stopped by heating and the acids were separated by silica gel chromatography (8). A sample of (+)-isocitrate was prepared similarly, although in most experiments DL-isocitrate was used.

Analytical methods. Aconitate was measured by its ultraviolet absorption; 240 $m\mu$ was the wavelength most frequently used. The molar extinction

coefficient varied with solvent, pH, and ionic strength. Citrate was measured by a chemical method (9) in earlier experiments. Subsequently we used an enzymic method involving the enzyme citratase (citrate oxaloacetate lyase, E.C.4.1.3.6) obtained from *Aerobacter aerogenes* grown in a medium containing citrate (10). Isocitrate was measured enzymically with isocitric dehydrogenase [L_s (sic)-isocitrate: NADP oxidoreductase, E.C.1.1.1.42] and NADP (11).

Assay procedure. Reaction mixtures generally consisted of 1 ml 0.1 M Tris-chloride buffer, varying concentrations of one of the three substrates, and 0.1 ml of appropriately diluted aconitase solution in 3 ml total volume; the final concentration of protein was usually 40–60 μg per milliliter. All experiments were carried out at 25°. The continuous formation or disappearance of aconitate was measured by following the change in ultraviolet absorption at 240 $m\mu$ on a recording chart attached to a Beckman DU spectrophotometer. In many experiments, after a short observation period (usually 3 minutes) the reaction was stopped and the mixture was analyzed for the three acids.

In some experiments coupled assays were used, consisting of either citrate, aconitase, isocitric dehydrogenase, Mn^{++} , and NADP; or isocitrate, aconitase, citratase, Mg^{++} , malic dehydrogenase,

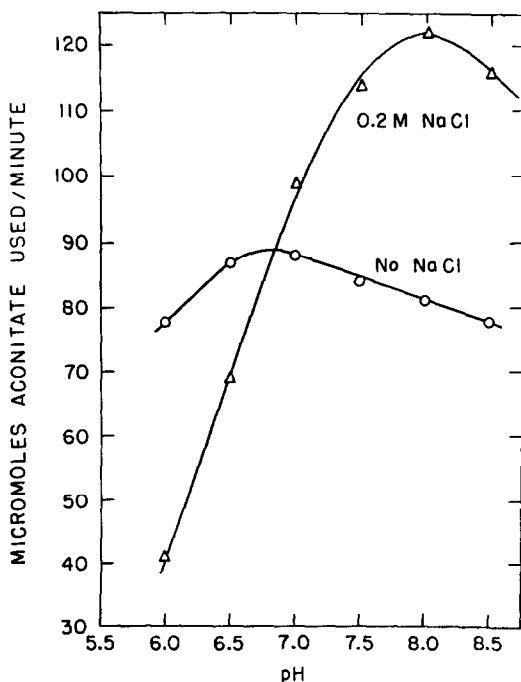


FIG. 1. Effect of ionic strength and pH on aconitase activity, based on rate of disappearance of aconitate. Each reaction mixture contained 0.02 M glycine and 0.0013 M *cis*-aconitate. The enzyme was prepared from beef heart.

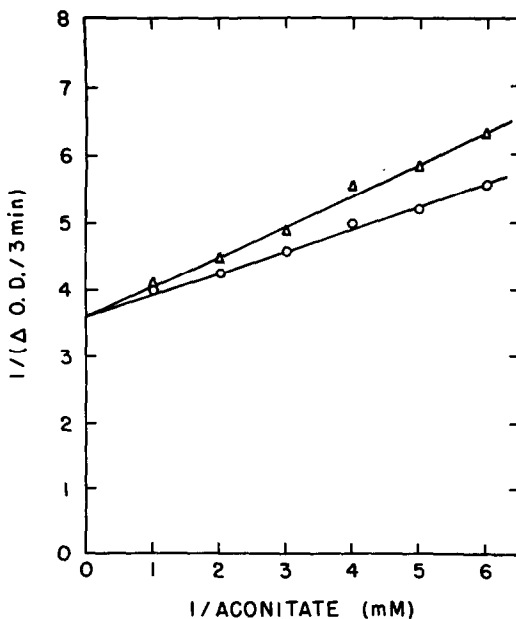


FIG. 2. Competitive inhibition of aconitase by *trans*-aconitate. (○) No *trans*-aconitate; (△) 0.003 M *trans*-aconitate added. Reaction mixtures contained 0.033 M Tris buffer, aconitase, and *cis*-aconitate as indicated.

TABLE I

EFFECT OF D₂O ON MICHAELIS CONSTANTS AND MAXIMUM VELOCITIES OF SUBSTRATES FOR ACONITASE

Substrate	H ₂ O	D ₂ O	D ₂ O/H ₂ O ratio
<i>cis</i> -Aconitate ^a			
V_{-A}	1.00	0.46	0.46
V_{AC}	0.53	0.14	0.26
V_{AI}	0.47	0.32	0.68
K_m	0.99×10^{-4} M (8) ^b	0.78×10^{-4} M (8)	0.79
Citrate			
V_{-C}	0.53	0.16	0.32
V_{CA}	0.25	0.09	0.39
V_{CI}	0.28	0.07	0.26
K_m	0.95×10^{-3} M (10)	0.85×10^{-3} M (7)	0.89
2-Deuteriocitrate			
V_{-C}	0.51	0.15	0.30
V_{CA}	0.24	0.09	0.36
V_{CI}	0.27	0.06	0.24
K_m	1.00×10^{-3} M (4)	0.94×10^{-3} M (3)	0.94
L(+)-Isocitrate			
V_{-I}	0.85	—	—
V_{IA}	0.41	—	—
V_{IC}	0.44	—	—
K_m	1.39×10^{-4} M (4)	—	—
L(+)-3-Deuterioisocitrate			
V_{-I}	0.81	0.22	0.27
V_{IA}	0.39	0.115	0.29
V_{IC}	0.42	0.105	0.25
K_m	1.27×10^{-4} M (4)	1.04×10^{-4} M (3)	0.82
DL-Isocitrate			
V_{-I}	0.82	0.22	0.26
V_{IA}	0.39	0.11	0.28
V_{IC}	0.43	0.11	0.25
K_m	1.48×10^{-4} M (10)	1.34×10^{-4} M (7)	0.90

^a V_{-A} is the maximum velocity for disappearance of *cis*-aconitate; V_{AC} and V_{AI} are maximum velocities of formation of citrate and isocitrate, respectively, from aconitate. Maximum velocities are given in arbitrary units; all values are relative to $V_{-A} = 1$.

^b Numbers in parentheses after K_m values indicate the numbers of experiments; in each experiment, duplicate assays of 6 substrate concentrations were usually employed.

and NADH₂. In these assays the increase or decrease, respectively, in absorption at 340 m μ was followed, the changes reflecting formation of isocitrate and citrate.

Kinetic analysis. Measurements were made, in both H₂O and D₂O, of the Michaelis constants of aconitate, citrate, deuteriocitrate, DL-isocitrate, (+)-isocitrate, and (+)-deuterioisocitrate, and of the maximum velocities for each possible interconversion among the substrates. In most experiments, 6 concentrations of substrate were used, and assays were performed in duplicate. Data were plotted according to the Lineweaver-Burk transformation, and regression lines were fitted by the method of least squares. The results of a number of experiments were averaged, reciprocals of

the standard errors of individual experiments serving as weighting coefficients.

RESULTS

Optimum pH and effects of ionic strength. Morrison showed that aconitase activity was profoundly influenced by the buffer system (12). We have confirmed and extended his observations, and have shown that the ionic strength of the assay system is of considerable importance. Figure 1 illustrates this point. With 0.02 M glycine-HCl buffer, the optimum pH in the absence of sodium chloride was about 7. In the presence of 0.2 M NaCl, however, there was

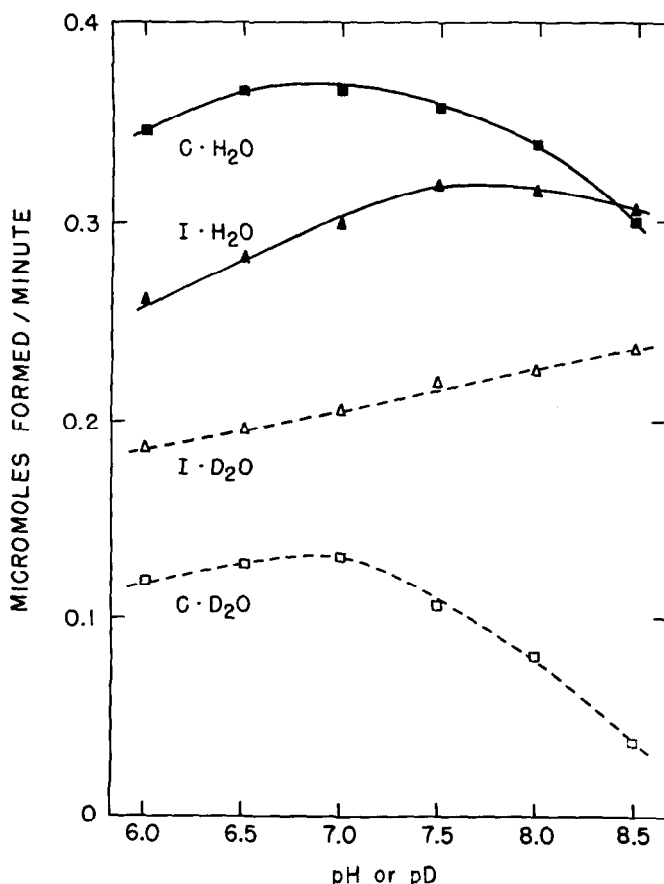


Fig. 3. Effect of pH and pD on formation of citrate and isocitrate from aconitate in H_2O and D_2O . Assay conditions were 0.003 M *cis*-aconitate and 0.016 M Tris + 0.016 M acetate buffer. Triangles, citrate formation; squares, isocitrate. Solid symbols and lines, H_2O ; open symbols and dotted lines, 99% D_2O . Analysis of citrate and isocitrate were carried out on samples deproteinized after a 3-minute incubation.

marked inhibition of activity below pH 6.5, but stimulation above pH 7; and the optimum pH was shifted from 7 to 8.

Somewhat different effects of sodium chloride were noted with other buffers. Since the pH optima for the three substrates are somewhat different, and since the activation effect of increased ionic strength was variable, we elected to carry out kinetic studies in 0.033 M Tris-chloride buffer, pH 7.4, with no electrolyte added.³ Assays in

³ In 0.033 M Tris buffer without added NaCl, the activity vs. pH curve is rather flat, and although the optimum is about 7.4, variations between 7 and 9 show relatively slight differences. The broadness of the pH optimum simplifies the problems of choice of pH and pD when comparing enzymic activities in H_2O and D_2O (13) (see Fig. 4).

99.6% D_2O were carried out at pD 7.4. The amount of exchangeable protium introduced into D_2O was calculated to be less than 0.5%, so that the final D_2O concentration was at least 99.1%.

Competitive inhibition by trans-aconitate. As indicated above, solutions of *cis*-aconitate contained about 6% of the *trans* isomer. We considered it important to establish how much this compound would affect the kinetic constants of *cis*-aconitate. Figure 2 shows the effect of 3×10^{-4} M *trans*-aconitate on the hydration of *cis*-aconitate. It can be calculated that in H_2O the inhibition constant, K_I , for *trans*-aconitate is 7.9×10^{-4} M, about 8 times the K_m for *cis*-aconi-

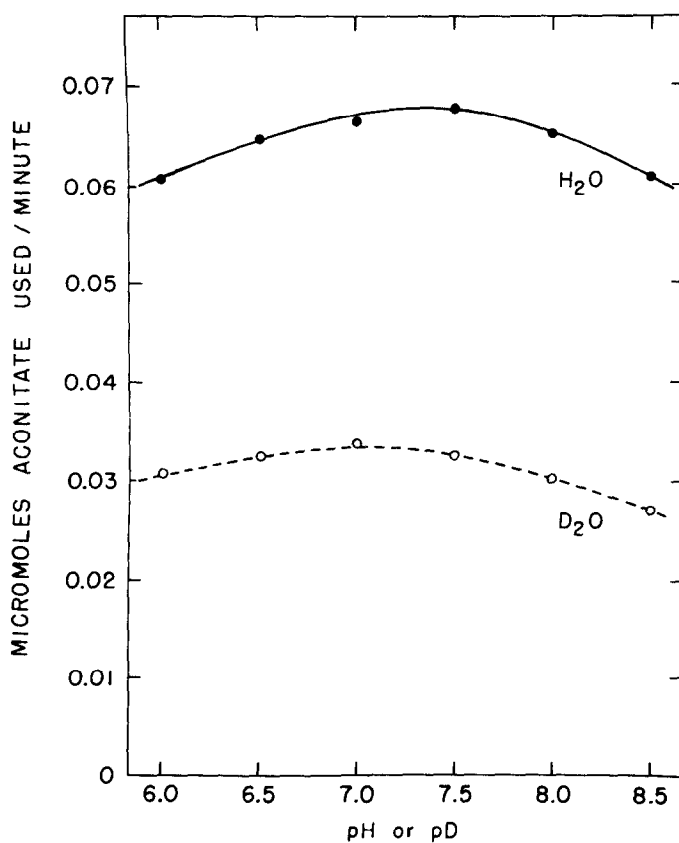


Fig. 4. Effect of pH and pD on disappearance of aconitate in H₂O and D₂O. Assay conditions were the same as in Fig. 3.

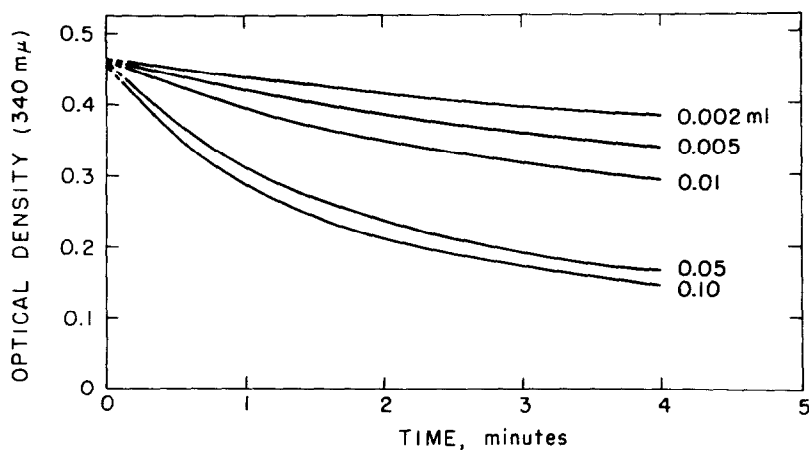


Fig. 5. Coupled assay of aconitase and citratase. Each reaction mixture contained 0.033 M Tris buffer, pH 7.4, 0.003 M MgSO₄, 0.0016 M DL-isocitrate, 7.6×10^{-5} M NADH, and 0.1 ml citratase solution. The reaction was started by adding the indicated amount of beef-heart aconitase solution. (Redrawn from chart tracings at 340 m μ .)

TABLE II
PERCENTAGE COMPOSITION OF ACONITASE EQUILIBRIUM MIXTURE IN H₂O AND D₂O

Composition	H ₂ O		D ₂ O	
	Observed ^a	Calculated ^b	Observed	Calculated ^c
Citrate	88.4 ± 1.3	87.7	87.7 ± 2.3	88.0
Isocitrate	7.5 ± 1.0	7.8	8.5 ± 1.2	9.0
Aconitate	4.1 ± 0.2	4.6	3.7 ± 0.4	3.0

^a Values for citrate and isocitrate are averages of 9 measurements (3 with each initial substrate), with standard deviations; values for aconitate are averages of 6 measurements (3 each with citrate and isocitrate).

^b Calculated from data in Table I for *cis*-aconitate, citrate, and L(+)-isocitrate in H₂O.

^c Calculated from data in Table I for *cis*-aconitate, 2-deuteriocitrate, and 3-deuterioisocitrate in D₂O.

tate.⁴ The *trans* isomer is less inhibitory in D₂O, with a K_I of 10.6×10^{-4} M.

Deuterium effects on kinetics. Values for the Michaelis constants and maximum velocities at pH or pD 7.4 are listed in Table I. It is apparent that there was considerable variation in the degree of inhibition of the various interconversions. One of the most striking effects was seen in the relative amounts of citrate and isocitrate formed from aconitate in D₂O. In ordinary water, the initial rates of formation of citrate and isocitrate are approximately the same; however, in D₂O the rate of formation of isocitrate is approximately twice that of citrate.

The Michaelis constants for all three substrates were appreciably lower in D₂O than in H₂O. It can be seen that in a given solvent the differences in both the Michaelis constants and the maximum velocities between (+)-isocitrate and 3-deuterioisocitrate and

between citrate and 2-deuteriocitrate were very slight. The unnatural isomer of isocitrate would seem to be a rather feeble competitive inhibitor; the K_I for threo-L₃-isocitrate was estimated to be about 4×10^{-3} M.

The Michaelis constant for *cis*-aconitate in H₂O is about the same as the values reported by Tomizawa (14) and Morrison (12); that for citrate is close to those obtained by Racker (15) and Tomizawa, but not Morrison, whereas the K_m for isocitrate is between one third and one half the values found by the other three authors. It should be pointed out that all of the investigators have used somewhat different experimental conditions; furthermore, some of Tomizawa's data (16) suggest the possibility of isozymes of aconitase from different sources.

The relative maximum velocities agree rather well with those of Morrison (5) except for the two interconversions between citrate and isocitrate; our values for V_{IC} and V_{CI} are about $2\frac{1}{2}$ and 5 times higher, respectively, than those of Morrison, based on the other four velocities.

Effect of pH and pD on citrate:isocitrate ratios. The observation that in D₂O the rate of isocitrate formation from aconitate was twice the rate of citrate formation prompted an investigation of pH effects on this ratio. The data in Fig. 3 show that in H₂O the percentage of aconitate converted to isocitrate increases from 43 to 51 between pH 6 and 8.5, whereas in D₂O the percentage increases from 61 to 86 over the same pD range. Furthermore, whereas the absolute rate of isocitrate formation showed a distinct

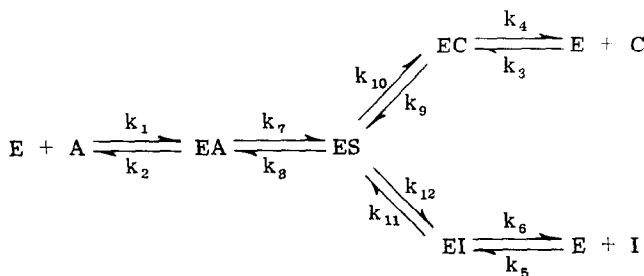
⁴ When a competitive inhibitor, I , is present as a fixed proportion, α , of the total substrate, S , containing true substrate and inhibitor, the rate equation becomes:

$$\frac{1}{v} = \frac{1}{V_m} \left(1 + \frac{\alpha}{1 - \alpha} \cdot \frac{K_s}{K_I} \right) + \frac{K_s}{V_m} \cdot \frac{1}{S} \cdot \frac{1}{1 - \alpha}$$

If a constant amount of the inhibitor is then added, the equation becomes:

$$\frac{1}{v} = \frac{1}{V_m} \left(1 + \frac{\alpha}{1 - \alpha} \cdot \frac{K_s}{K_I} \right) + \frac{K_s}{V_m} \cdot \frac{1}{S} \cdot \frac{1}{1 - \alpha} \left(1 + \frac{I}{K_I} \right)$$

Thus the ratio of the slopes with and without the added inhibitor gives $1 + I/K_I$, regardless of the fraction of impurity present in the substrate.



SCHEME I

maximum in H_2O at about pH 7.5, there was no maximum observed in D_2O over the range of pD studied. Figure 4 shows the data on aconitate disappearance in this experiment.

Coupled assays. Among the mechanisms which have been proposed for aconitase are the "linear," "tricyclic," and "centrosymmetric" (17). According to the linear model, *cis*-aconitate is an obligatory intermediate between citrate and isocitrate, whereas the other two models postulate an interconversion between the aconitase-citrate and aconitase-isocitrate complexes without the formation of free aconitate. If the linear mechanism is correct, there should be a lag period in the formation of citrate from isocitrate (and *vice versa*) (17). Thus if one prepared an incubation mixture of isocitrate, citratase, malic dehydrogenase, Mg^{++} , and NADH_2 , a delay in the oxidation of NADH (measured at 340 $\text{m}\mu$) after addition of aconitase would be evidence for the linear mechanism. The data in Fig. 5 show that no lag was detectable. Similar observations were made when citrate was used as the substrate and the aconitase reaction was coupled with isocitric dehydrogenase.

Deuterium effects on equilibrium. In several experiments the incubations were carried out until equilibrium was reached. The composition of the equilibrium mixture was then determined. Table II shows that the principal difference between equilibrium conditions in H_2O and those in D_2O was a slightly higher percentage of isocitrate in D_2O .

The predicted values were calculated from the data in Table I according to the procedure described in the next section.

Relationship between equilibrium and kinetics. It is possible to check the internal consistency of kinetic data for an enzymic reaction by using them to calculate the equilibrium constant for the reaction. The so-called "Haldane relation" for aconitase was derived as follows.

For the simplest form of the mechanism involving a common intermediate (3). (See Scheme I) where E , A , C , and I represent the concentrations of free enzyme, *cis*-aconitate, citrate, and isocitrate, respectively; EA , EC , and EI the corresponding enzyme-substrate complexes, and ES the common intermediate, the following rate equations are obtained by the steady-state treatment:

$$\begin{aligned}
 -\frac{dA}{dt} &= E_0 \left(\frac{k_1 k_7 [k_4 k_{10} (k_6 + k_{11}) + k_6 k_{12} (k_4 + k_9)] A - k_2 k_3 k_8 k_9 (k_6 + k_{11}) C - k_2 k_5 k_8 k_{11} (k_4 + k_9) I}{D_0 + k_1 D_A A + k_3 D_C C + k_5 D_I I} \right) \\
 -\frac{dC}{dt} &= E_0 \left(\frac{k_3 k_9 [k_2 k_8 (k_6 + k_{11}) + k_6 k_{12} (k_2 + k_7)] C - k_1 k_4 k_7 k_{10} (k_6 + k_{11}) A - k_4 k_5 k_{10} k_{11} (k_2 + k_7) I}{D_0 + k_1 D_A A + k_3 D_C C + k_5 D_I I} \right) \\
 -\frac{dI}{dt} &= E_0 \left(\frac{k_5 k_{11} [k_4 k_{10} (k_2 + k_7) + k_2 k_8 (k_4 + k_9)] I - k_1 k_6 k_7 k_{12} (k_4 + k_9) A - k_3 k_6 k_9 k_{12} (k_2 + k_7) C}{D_0 + k_1 D_A A + k_3 D_C C + k_5 D_I I} \right),
 \end{aligned}$$

where

$$\begin{aligned}
 D_0 &= k_2k_4k_8k_{10} + k_2k_4k_6k_{10} + k_2k_4k_6k_{12} \\
 &\quad + k_2k_4k_8k_{11} + k_2k_4k_{10}k_{11} + k_2k_6k_8k_9 \\
 &\quad + k_2k_6k_9k_{12} + k_2k_8k_9k_{11} + k_4k_6k_7k_{10} \\
 &\quad + k_4k_6k_7k_{12} + k_4k_7k_{10}k_{11} + k_6k_7k_9k_{12} \\
 D_A &= k_4k_6k_7 + k_4k_6k_8 + k_4k_6k_{10} + k_4k_6k_{12} \\
 &\quad + k_4k_7k_{11} + k_4k_7k_{12} + k_4k_8k_{11} + k_4k_{10}k_{11} \\
 &\quad + k_6k_7k_9 + k_6k_7k_{10} + k_6k_8k_9 + k_6k_9k_{12} \\
 &\quad + k_7k_9k_{11} + k_7k_9k_{12} + k_7k_{10}k_{11} \\
 &\quad + k_8k_9k_{11} \\
 D_C &= k_2k_6k_8 + k_2k_6k_9 + k_2k_6k_{10} + k_2k_6k_{12} \\
 &\quad + k_2k_8k_{11} + k_2k_9k_{11} + k_2k_9k_{12} \\
 &\quad + k_2k_{10}k_{11} + k_6k_7k_9 + k_6k_7k_{10} \\
 &\quad + k_6k_7k_{12} + k_6k_8k_9 + k_7k_9k_{11} \\
 &\quad + k_7k_9k_{12} + k_7k_{10}k_{11} + k_8k_9k_{11} \\
 D_I &= k_2k_4k_8 + k_2k_4k_{10} + k_2k_4k_{11} + k_2k_4k_{12} \\
 &\quad + k_2k_6k_9 + k_2k_6k_{11} + k_2k_6k_{12} + k_2k_{10}k_{11} \\
 &\quad + k_4k_7k_{10} + k_4k_7k_{11} + k_4k_7k_{12} + k_4k_8k_{11} \\
 &\quad + k_7k_9k_{11} + k_7k_9k_{12} + k_7k_{10}k_{11} \\
 &\quad + k_8k_9k_{11}.
 \end{aligned}$$

Under initial conditions in which two of the three substrates are absent, the equations reduce to the form of the Michaelis-Menten equation:

$$v = V_{\max}S/(S + K_m),$$

so that the maximum velocities and Michaelis constants can be defined as follows for this particular mechanism:⁵

$$\begin{aligned}
 V_{AC} &= \frac{k_4k_7k_{10}(k_8 + k_{11})}{D_A} \cdot E_0 & V_{AI} &= \frac{k_6k_7k_{12}(k_4 + k_9)}{D_A} \cdot E_0 \\
 V_{CA} &= \frac{k_2k_8k_9(k_6 + k_{11})}{D_C} \cdot E_0 & V_{CI} &= \frac{k_6k_9k_{12}(k_2 + k_7)}{D_C} \cdot E_0 \\
 V_{IA} &= \frac{k_2k_8k_{11}(k_4 + k_9)}{D_I} \cdot E_0 & V_{IC} &= \frac{k_4k_{10}k_{11}(k_2 + k_7)}{D_I} \cdot E_0 \\
 K_A &= D_0/k_1D_A & K_C &= D_0/k_3D_C & K_I &= D_0/k_5D_I \\
 V_A &= V_{AC} + V_{AI} & V_C &= V_{CA} + V_{CI} & V_I &= V_{IA} + V_{IC}
 \end{aligned}$$

⁵ It is immaterial whether a centrosymmetric or tricyclic mechanism (17) is assumed, since the rate equations for both reduce to the same form. Similarly the introduction of additional intermediate complexes does not change the final equations. The significance of the K and V constants, of course, will depend on the reaction mechanism.

However, for the linear mechanism involving free aconitate as an intermediate between citrate and isocitrate

The original rate equations can then be written:

$$\begin{aligned}
 -\frac{dA}{dt} &= \frac{AV_A/K_A - CV_{CA}/K_C - IV_{IA}/K_I}{1 + A/K_A + C/K_C + I/K_I} \\
 -\frac{dC}{dt} &= \frac{CV_C/K_C - AV_{AC}/K_A - IV_{IC}/K_I}{1 + A/K_A + C/K_C + I/K_I} \\
 -\frac{dI}{dt} &= \frac{IV_I/K_I - AV_{AI}/K_A - CV_{CI}/K_C}{1 + A/K_A + C/K_C + I/K_I}.
 \end{aligned}$$

At equilibrium, the rate equations then become:

$$\begin{aligned}
 V_AA/K_A &= V_{CA}C/K_C + V_{IA}I/K_I \\
 V_C C/K_C &= V_{AC}A/K_A + V_{IC}I/K_I \\
 V_I I/K_I &= V_{AI}A/K_A + V_{CI}I/K_I.
 \end{aligned}$$

Elimination of I from the equations yields:

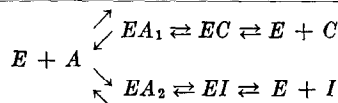
$$\frac{A}{C} = \frac{K_A}{K_C} \frac{V_{CA}V_{IA} + V_{CA}V_{IC} + V_{IA}V_{CI}}{\bar{V}_{AC}\bar{V}_{IA} + V_{AC}V_{IC} + V_{AI}\bar{V}_{IC}}. \quad (1)$$

Similarly,

$$\frac{I}{C} = \frac{K_I}{K_C} \frac{V_{CI}V_{AC} + V_{CI}V_{AI} + V_{CA}V_{AI}}{\bar{V}_{AC}\bar{V}_{IA} + V_{AC}V_{IC} + V_{IA}\bar{V}_{IC}}. \quad (2)$$

$$\frac{A}{I} = \frac{K_A}{K_C} \frac{V_{CA}V_{IA} + V_{CA}V_{IC} + V_{IA}V_{CI}}{\bar{V}_{CI}\bar{V}_{AC} + V_{CI}V_{AI} + V_{CA}\bar{V}_{AI}}. \quad (3)$$

The agreement between the calculated and observed results (Table II) in H_2O is excellent. In D_2O , however, the percentage of aconitate predicted from the kinetic data is too low by about 20%, although the ratio of citrate to isocitrate agrees fairly



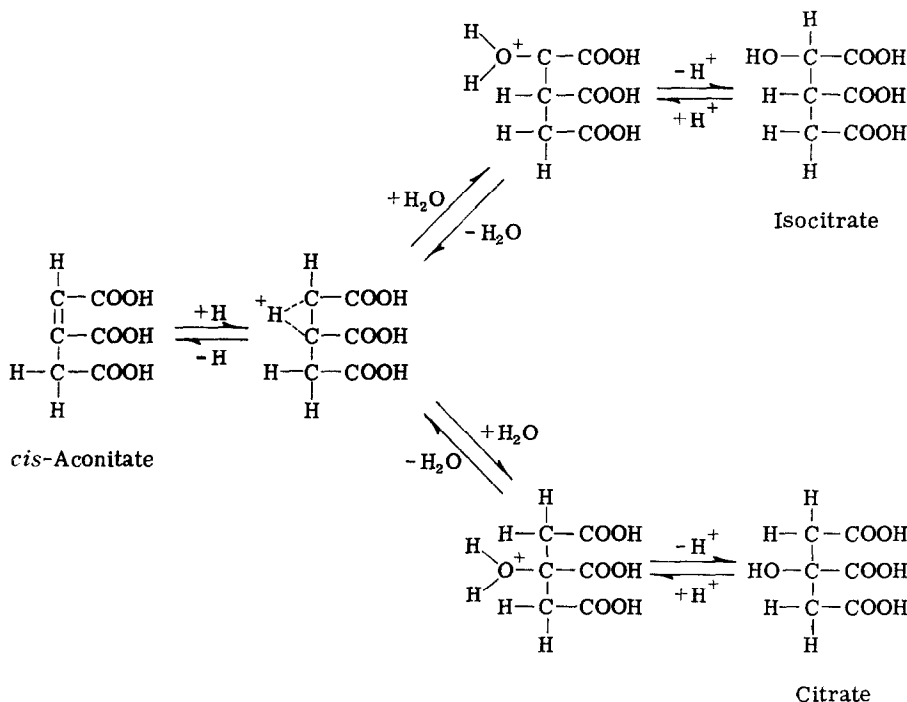
the equilibrium conditions are

$$\frac{A}{C} = \frac{K_A V_C}{K_C V_{AC}} \quad \text{and} \quad \frac{I}{C} = \frac{K_I V_C V_{AI}}{K_C V_I V_{AC}}.$$

Substitution of the appropriate constants yields a predicted equilibrium of 84.4:6.8:8.8 for citrate:isocitrate:aconitate in H_2O .

well with the observed value. The source of the discrepancy is probably the value for the Michaelis constant for aconitate in D_2O , for which a large standard error (nearly 30% of the value) was observed.

The data thus substantially confirm the centrosymmetric mechanism proposed by Speyer and Dickman (3), as modified by England and Colowick (18) and Gawron *et al.* (19):



SCHEME II

Substitution of the data of Morrison (12) into the above equations yields a predicted equilibrium of 95.3:3:8:0.9 for citrate:isocitrate:aconitate.

DISCUSSION

Aside from the solvent effects on the reaction rates shown in Table I, the most interesting observations are those which show a nearly negligible primary isotope effect with deuteriated citrate and isocitrate.

The fact that a very slight isotope effect was noted suggests that in the dehydration of citrate and isocitrate the first step, which is also rate-limiting, is the removal of the hydroxyl rather than the hydrogen. If the hydrogen removal were rate-limiting, there should have been a sizable isotope effect with deuteriocitrate and deuterioisocitrate. The data of England and Colowick (18) seem to rule out the possibility that a rapid splitting of a C—H bond precedes a slow rupture of a C—O bond.

A steady-state equation describing this mechanism would require formulation in terms of at least 16 rate constants (5 pairs for the reactions shown above and 3 more for the formation of enzyme-substrate complexes); even the simplest mechanism involving a common intermediate requires 6 pairs of rate constants. Since conventional kinetic analysis can yield only 9 independent values related to these rate constants, evaluation of individual rate constants is impossible. However, it can be predicted that since the solvent effects on the Michaelis constants are less than those on the maximum velocities, the effect will be normal throughout the range of substrate concentrations.

It appears that the mechanisms for aconitase and fumarase (20, 21) are similar, despite the fact that the former hydrates a molecule of the *cis*-configuration and the latter a *trans*-configuration. One point of difference is the much greater inhibition by

D₂O of the dehydration of citrate and isocitrate in comparison with malate (1), despite the fact that in both cases the rate-limiting step is the rupture of a C—O bond.

An observation for which we have no adequate explanation is the preferential formation of isocitrate from aconitate in D₂O. The difference in molecular dimensions of D₂O and H₂O (less than 1%) would seem to be too small to produce steric effects. We have repeatedly confirmed this observation, and the kinetic effects are reflected by the increased percentage of isocitrate at equilibrium in D₂O.

According to Krebs, the equilibrium conditions for aconitase are 90.9% citrate, 6.2% isocitrate, and 2.9% aconitate at 25°, pH 7.4 (22). These measurements were made with large quantities (3 gm/100 ml) of a crude extract of sheep heart; isocitrate and aconitate were estimated in ether extracts of the deproteinized incubation media, and citrate was calculated by difference. We believe that our procedures introduce fewer sources of error such as loss by adsorption or incomplete extraction, and we suggest that the higher percentages of isocitrate and aconitate reported here may be more nearly correct.

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